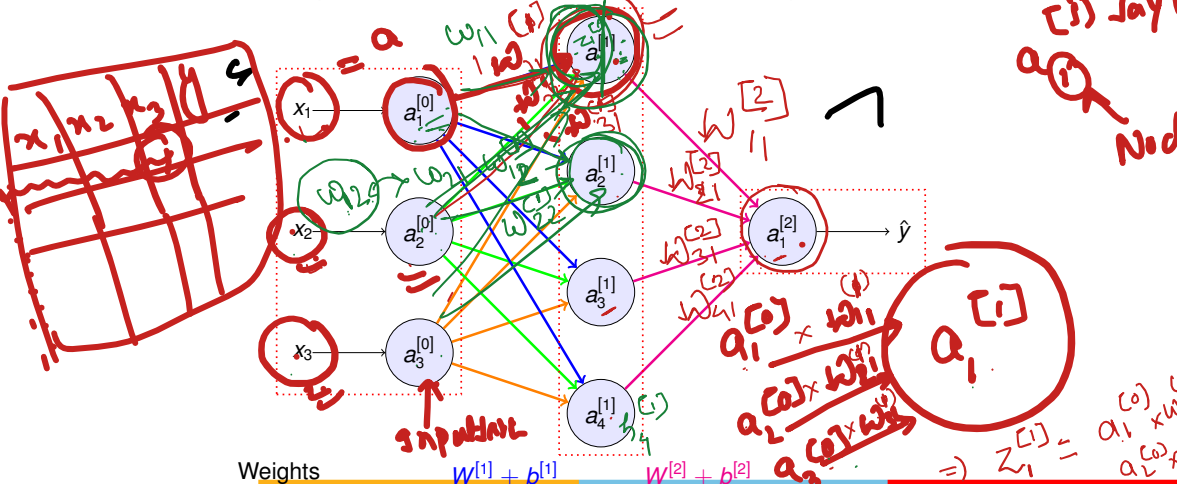


The diagram illustrates a neural network architecture with three layers: Input Layer, Hidden Layer, and Output Layer. The layers are represented by colored bars: yellow for the Input Layer, blue for the Hidden Layer, and red for the Output Layer. The labels are positioned below their respective bars.



COMPUTING THE ACTIVATIONS

For Hidden layer

$a^{[1]} = 6$ in the matrix form

$$\begin{aligned} h_1 \{ \begin{aligned} z_1^{[1]} &= w_1^{[1]} a^{[0]} + b^{[1]} \\ a_1^{[1]} &= \sigma(z_1^{[1]}) \end{aligned} \\ h_2 \{ \begin{aligned} z_2^{[1]} &= w_2^{[1]} a^{[0]} + b^{[1]} \\ a_2^{[1]} &= \sigma(z_2^{[1]}) \end{aligned} \\ h_3 \{ \begin{aligned} z_3^{[1]} &= w_3^{[1]} a^{[0]} + b^{[1]} \\ a_3^{[1]} &= \sigma(z_3^{[1]}) \end{aligned} \\ z_4^{[1]} &= w_4^{[1]} a^{[0]} + b^{[1]} \\ a_4^{[1]} &= \sigma(z_4^{[1]}) \end{aligned}$$

$$\begin{bmatrix} z_1^{[1]} \\ z_2^{[1]} \\ z_3^{[1]} \\ z_4^{[1]} \end{bmatrix} = \underbrace{\begin{bmatrix} \dots & w_1^{[1]T} & \dots \\ \dots & w_2^{[1]T} & \dots \\ \dots & w_3^{[1]T} & \dots \\ \dots & w_4^{[1]T} & \dots \end{bmatrix}}_{4 \times 3} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{3 \times 1} + \underbrace{\begin{bmatrix} b_1^{[1]} \\ b_2^{[1]} \\ b_3^{[1]} \\ b_4^{[1]} \end{bmatrix}}_{4 \times 1}$$

$$\begin{bmatrix} w_1^{[1]T} a^{[0]} + b_1^{[1]} \\ w_2^{[1]T} a^{[0]} + b_2^{[1]} \\ w_3^{[1]T} a^{[0]} + b_3^{[1]} \\ w_4^{[1]T} a^{[0]} + b_4^{[1]} \end{bmatrix} \leftarrow 4 \times 1 \text{ matrix}$$

$$a^{[1]} = \sigma(z^{[1]}) \leftarrow 4 \times 1 \text{ matrix}$$

$$z^{[0]} = \dots$$

ACTIVATIONS FOR TRAINING EXAMPLES

Vectorizing for one training example

$$\begin{aligned}
 a^{[0]} &= X \\
 \left\{ \begin{aligned}
 z^{[1]} &= W^{[1]} a^{[0]} + b^{[1]} \\
 a^{[1]} &= \sigma(z^{[1]})
 \end{aligned} \right. \\
 z^{[2]} &= W^{[2]} a^{[1]} + b^{[2]} \\
 a^{[2]} &= \sigma(z^{[2]}) \\
 \hat{y} &= a^{[2]}
 \end{aligned}$$



Vectorizing for all training examples

$$\text{Example 1 : } X^{(1)} \longrightarrow a^{[2](1)} = \hat{y}^{(1)}$$

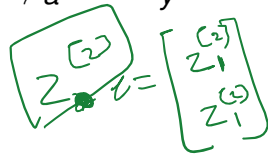
$$\text{Example 2 : } X^{(2)} \longrightarrow a^{2} = \hat{y}^{(2)}$$

$$\text{Example 3 : } X^{(2)} \longrightarrow a^{[2](3)} = \hat{y}^{(3)}$$

$\vdots$

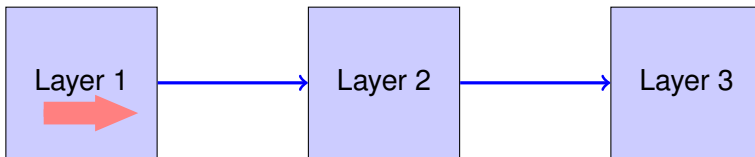
$\vdots$

$$\text{Example m : } X^{(m)} \longrightarrow a^{[2](m)} = \hat{y}^{(m)}$$



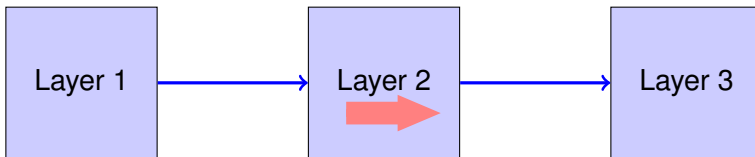
NEURAL NETWORK TRAINING - FORWARD PASS

Step 1: Compute Loss on mini-batch [Forward pass]



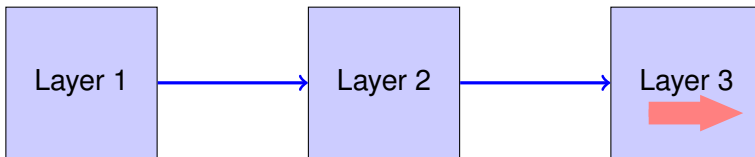
NEURAL NETWORK TRAINING - FORWARD PASS

Step 1: Compute Loss on mini-batch [Forward pass]



NEURAL NETWORK TRAINING - FORWARD PASS

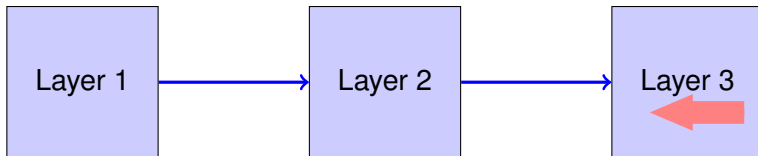
Step 1: Compute Loss on mini-batch [Forward pass]



NEURAL NETWORK TRAINING - BACKWARD PASS

Step 1: Compute Loss on mini-batch [Forward pass]

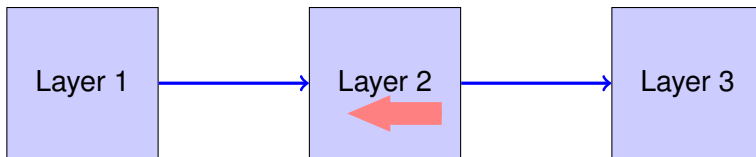
Step 2: Compute gradients wrt parameters [Backward pass]



NEURAL NETWORK TRAINING - BACKWARD PASS

Step 1: Compute Loss on mini-batch [Forward pass]

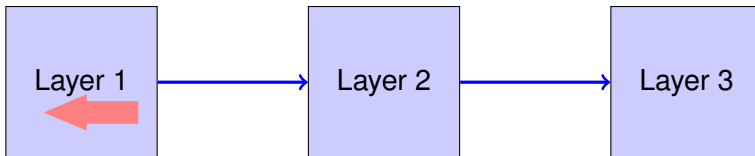
Step 2: Compute gradients wrt parameters [Backward pass]



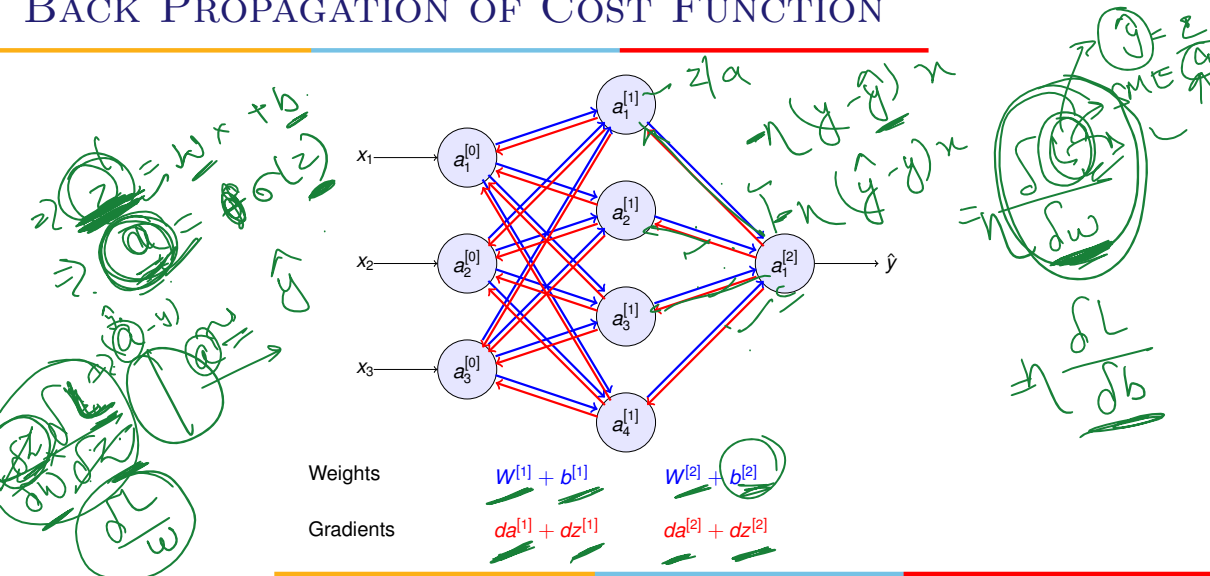
NEURAL NETWORK TRAINING - BACKWARD PASS

Step 1: Compute Loss on mini-batch [Forward pass]

Step 2: Compute gradients wrt parameters [Backward pass]



BACK PROPAGATION OF COST FUNCTION



DERIVATION OF GRADIENTS IN LAST LAYER

$$z = w^T x + b$$

$$a = \sigma(z)$$

$$L = (-y \log a) - (1 - y) \log(1 - a)$$

$$\frac{\partial L}{\partial a} = \frac{-y}{a} + \frac{1 - y}{1 - a}$$

$$dz = \frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} = \left(\frac{-y}{a} + \frac{1 - y}{1 - a} \right) a(1 - a) = a - y$$

$$dw = \frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial w} = (a - y)x$$

COMPUTING THE GRADIENTS

For all training examples

$$dZ^{[2]} = A^{[2]} - Y$$

$$dW^{[2]} = \frac{1}{m} (dZ^{[2]} \cdot A^{[1]T})$$

$$db^{[2]} = \frac{1}{m} \sum dZ^{[2]}$$

$$dZ^{[1]} = (W^{[2]T} dZ^{[2]}) * \sigma'(Z^{[1]})$$

$$dW^{[1]} = \frac{1}{m} (dZ^{[1]} \cdot X^T)$$

$$db^{[1]} = \frac{1}{m} \sum dZ^{[1]}$$

For one training example

$$dz^{[2]} = a^{[2]} - y$$

$$dw^{[2]} = dz^{[2]} \cdot a^{[1]T}$$

$$db^{[2]} = dz^{[2]}$$

$$dz^{[1]} = (w^{[2]T} dz^{[2]}) * \sigma'(z^{[1]})$$

$$dw^{[1]} = dz^{[1]} \cdot x^T$$

$$db^{[1]} = dz^{[1]}$$